

Chapter-01

1) Distinguish between computer architecture and organizations (Q-2018, 17, -)

-02

Ans: Computer architecture refers to those attributes of a system visible to a programmer. Or, those attributes that have a direct impact on the logical execution of a program. Computer organization refers to the operational units and their interconnections that realize the architectural specifications. As example, architectural attributes include the instruction set, I/O mechanisms, the number of bits used to represent various data types and techniques for addressing memory. Organizational attributes include control signals, interfaces between the computer and peripherals and the memory technology used.

(2) Why do we need to study computer organization and architecture? — 02

Ans: The computer lies at the heart of computing. Without it most of the computing disciplines today would be a branch of theoretical mathematics. To be a professional in any field of computing today, one should not regard a computer as a black box that executes programs by magic. All students of computing should acquire some understanding of a computer system's functional unit, their characteristics, their performance and their interactions.

A good overall understanding of these concepts will be useful in other areas of study and in future work.

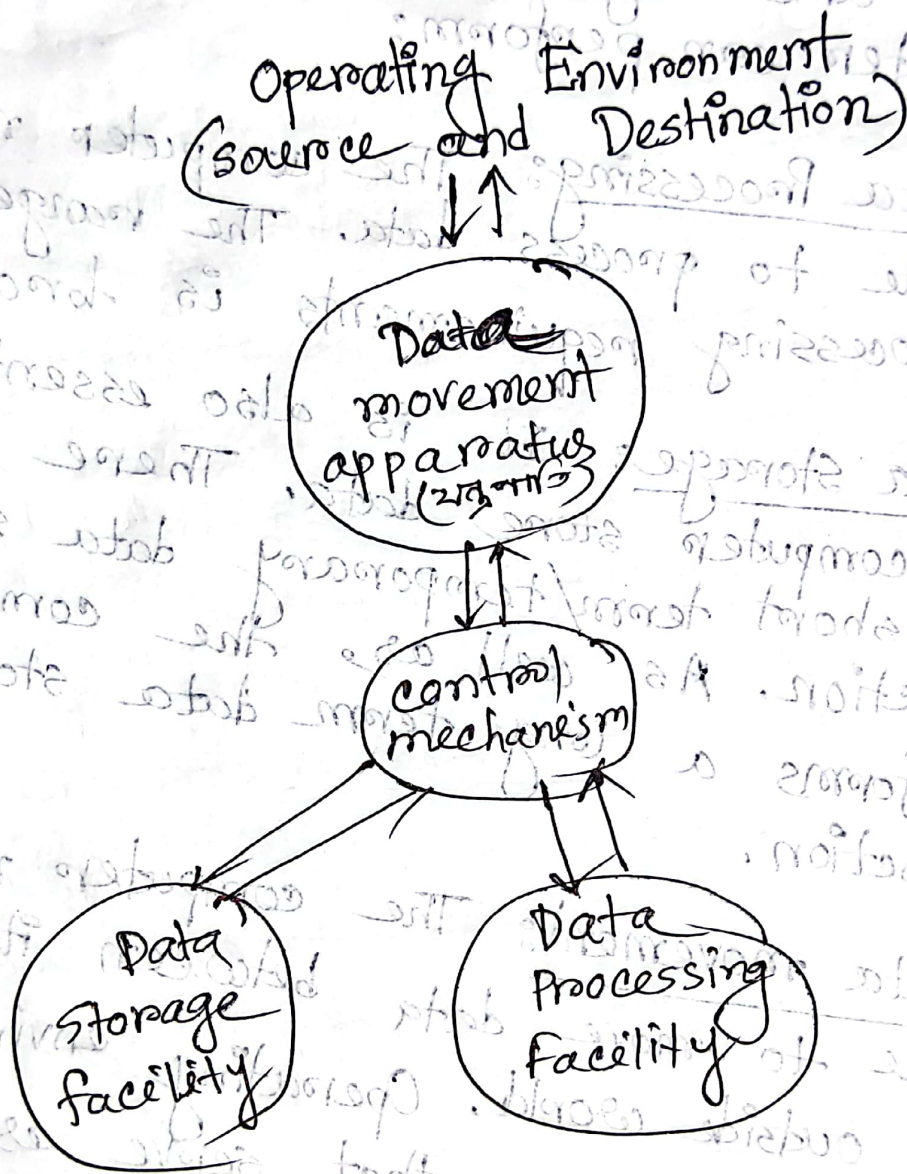
(3) Explain the basic functions of a computer (4)

Ans:

There are only four basic functions that a computer can perform:

- (i) Data Processing: The computer must be able to process data. The range of processing requirements is broad.
- (ii) Data Storage: It is also essential that a computer store data. There is at least a short term/temporary data storage function. As well as, the computer performs a long term data storage function.
- (iii) Data movement: The computer must be able to move data between itself and the outside world. Operating environment consists of devices that serve as either source or destinations of data. When data are received from or delivered to a device directly connected to the computer, known as I/O, device is called peripheral. If distance is longer process is known as data communication.

(iv) Control: There must be control of these three functions.



(4) Establish a relationship between computer architecture and organization.

Ans: २ नं प्रश्न उत्तर +

In microcomputers, the relationship between architecture and organization is very close. Changes in technology not only influence organization but also architecture. There is less of a requirement for generation-to-generation compatibility for these smaller machines. Thus, there is more interplay between organization and architectural design decisions.

(5) Explain structures of computer (कम्प्यूटर की संरचना)

Ans: There are four main structural components:

- i) Central Processing Unit (CPU)
- ii) Main memory
- iii) I/O
- iv) system interconnection.

Components of CPU — Control Unit, ALU, Registers, CPU interconnection.

Chapter — 11

① Mention the advantages and disadvantages of variable length addressing. — 2

** অসম্পূর্ণ তথ্য
* তাই প্রাথমিক লক্ষ্য ছিল " = বই ৪০০ পৃষ্ঠা

② Describe various addressing mode of instruction set — 3

বই ৩৪৩ পৃষ্ঠা !!

X বাদ দিচ্ছে দিচ্ছি

Chapter 12

1) What general roles are performed by processor registers? — 2

Ans: The registers in CPU perform two roles:

i) User-visible Registers: These enable the machine or assembly language programmer to minimize main memory references by optimizing use of registers.

ii) Control and Status Registers: These are used by the control unit to control the operation of the CPU and operating system programs to control the execution of programs.

② ~~i) Pipeline processor is insufficient for branch instruction. Explain the mechanism to deal the branches in pipeline.~~

ii) Draw a timing diagram for instruction pipeline for a sequence of 9 instructions in which the fifth instr. is a branch that is taken, no data dependencies

~~iii) Show that, Pipeline processor with k stages is k time faster than non pipeline processor.~~

*** iii) ~~ଆମର ସମସ୍ତ — ମାରିମଲାଇଁ କି ? ବିଭାବ~~
performance ବାଢ଼ାଉଁ କି ?
Disadvantages

Ans:

Page - 424 (m...)

Time diagram
12/10, 12/11

Pipeline:

Pipeline is similar to the use of an assembly line in a manufacturing plant. An assembly line takes advantage of the fact that a product goes through various stages of production. Products on various stages can be worked on simultaneously. This process is also referred as pipelining, because, new inputs are accepted at one end before previously accepted inputs appear as output at the other end.

Pipelining is a process of arrangement of hardware elements of the CPU such that its overall performance is increased.

Has two independent stages:

i) First stage - Fetches an instruction and buffers it.

ii) Second stage - ~~Free~~. When the second stage is free, the first stage passes it the buffers instruction.

While the second stage is executing the instruction, the first stage takes advantage of any unused memory cycles to fetch and buffer the next instruction. This is called instruction fetches or fetch overlap.

* Performance →

Let us consider the cycle time of an instruction pipeline is τ . The cycle time can be determined as,

$$\tau = \max[\tau_i] + d = \tau_{\text{im}} + d, \quad 1 \leq i \leq K$$

where,

τ_{im} = maximum stage delay

K = number of stages in the instruction pipeline.

d = time delay of ~~latch~~ latch, needed to advance signals and data from one stage to the next.

In general, the time delay d is equivalent to a clock pulse and $\tau_{\text{im}} \gg d$. Now, if n instructions are processed with no branches,

the total time required T_K to execute all instructions is,

$$T_K = [K + (n-1)]T$$

The ninth instruction completes at time cycle 14:

$$14 = [6 + (9-1)]$$

The speedup factor for the instruction pipeline compared to execution without the pipeline is defined as:

$$S_K = \frac{T_1}{T_K} = \frac{nKT}{[K + (n-1)]T} = \frac{nK}{K + (n-1)}$$

The larger the number of pipeline stages, the greater the potential for speedup.

Disadvantages:

- If looping happens, the performance will not be better.
- Its complexity is a disadvantage also.
- Inability to constantly run at full speed.

The Pentium Processor [disadvantages]

The register organization includes the following types of registers -

i) General: There are 8 32-bit general purpose registers. Used for all type of pentium instructions.

ii) Segment: The six 16-bit segment registers contain segment selectors.

iii) Flags: The EFlags register contains condition codes and various mode bits.

iv) Instruction Pointer: Contains the address of the current instruction.

v) Numeric: There are 8 registers that function as a stack, holds 80-bit floating point numbers.

vi) Control: The 16 bit control register contains bits the control the operation of floating point unit.

vii) Status: The 16 bit status register contains bits that reflect the current state of the floating point unit.

vii) Tag Word: 16 bit register contains a 2-bit tag for each floating point.

viii) Data Pointer

Chapter - 10

1) * Elements of a machine instructions

The elements of a machine instructions are as follows :-

(i) operation code : Specifies the operation to be performed like ADD or I/O.

The operation is specified by a binary code, known as the operation code or opcode.

(ii) Source Operand References: The operation may involve one or more source operands, that is, operands that are inputs for the operation.

(iii) Result Operand Reference: The operation may produce result.

(iv) Next instruction Reference: This tells the CPU where to fetch the next instruction after the execution of this instruction is complete.

2) Any program written in a high-level language must be translated into machine language to be executed. With this in mind, we can categorize instruction types as follows:

- i) Data Processing: Arithmetic or logic instruction.
- ii) Data Storage: Memory Instructions.
- iii) Data Movement: I/O instructions.
- iv) Control: Test and branch instructions.

3) Instruction Set Design:

The instruction set design is very complex, because it affects so many aspects of computer system. It defines many of the functions performed by the CPU. Its like controlling the CPU. The most important of these fundamental design issues include the following:

- i) Operation Repertoire: How many and which operations to provide, how complex operations should be.
- ii) Data types: The various types of data upon which operations are performed.

(iii) Instruction Format: Instruction length, number of addresses, size of various fields and so on.

(iv) Registers: Number of CPU registers that can be referenced by instructions and their use.

(v) Addressing: The mode or modes by which the address of an operand is specified.

4) * Types of Operands:

Machine instructions operate on data.

The most important general categories are:-

~~(i) Addresses~~

~~(ii) Numbers~~

~~(iii) Characters~~

~~(iv) Logical Data~~

(i) Addresses: Addresses are a form of data. In many cases, some calculation must be performed on the operand reference in an instruction to determine the main or virtual memory.

(i) Numbers: All machine languages include numeric data types. Three types of numerical data are common in computers — Integer or fixed point, Floating point, decimal.

(ii) Characters: A common form of data is text or character strings.

(v) Logical Data: Each word or other addressable unit like byte, halfword and so on are treated as a single unit of data. To consider an n bit unit as consisting of n 1 bit items of data, each item having the value of 0 or 1. When data are viewed this way, they are considered to be logical data.

5) Pentium Data Types:

The pentium can deal with data types of 8, 16, 32 and 64 bits in length. There are several pentium data types: —

From Book — 339 page 13 (13)

6) Types of Operations:

The number of different opcodes varies from machine to machine. A useful and typical categorization is:

(i) Data Transfer.

(ii) Arithmetic

(iii) Logical

(iv) Conversion

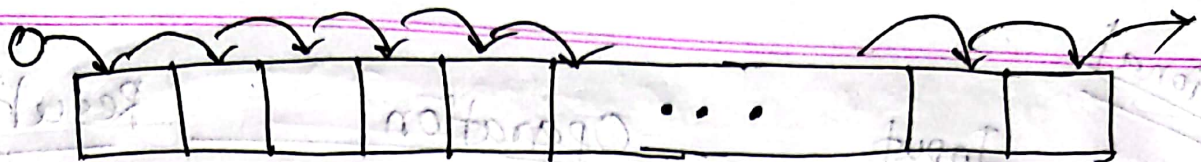
(v) I/O

(vi) System control

342 प्रश्न
subcategory

7) Logical shift and Arithmetic Shifts

With a logical shift, the bits of a word are shifted left or right. On one end, the bit shifted out is lost. On the other end, a 0 is shifted in. Logical shifts are useful primarily for isolating fields within a word. The 0s that are shifted into a word displace unwanted information that is shifted off the other hand.

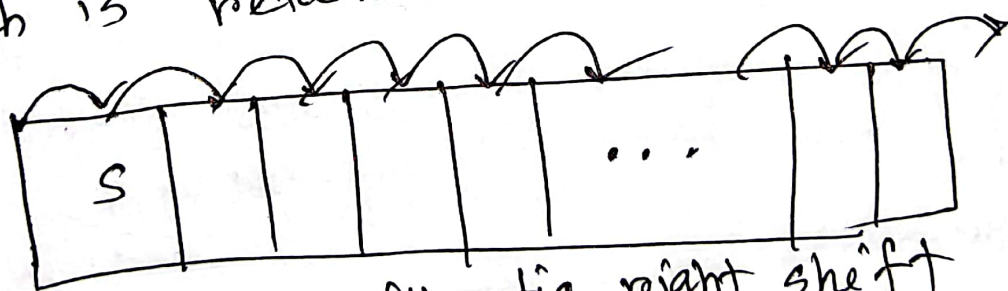


(a) Logical Right shift



(b) Logical Left shift

The arithmetic shift operation treats the data as a signed integer and does not shift the sign bit. On a right arithmetic shift, the sign bit is replicated into the bit position to its right. On a left arithmetic shift, a logical left shift is performed on all bits but sign bit, which is retained.



Arithmetic right shift

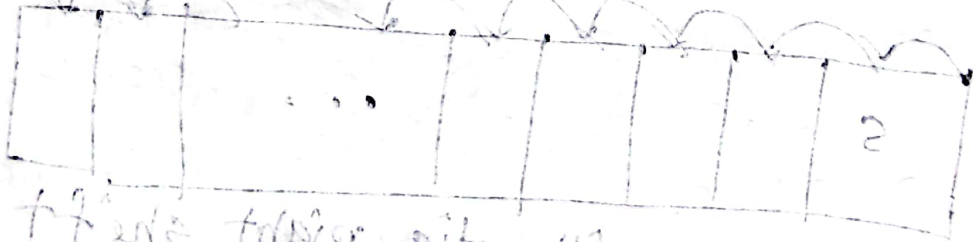


Arithmetic left shift

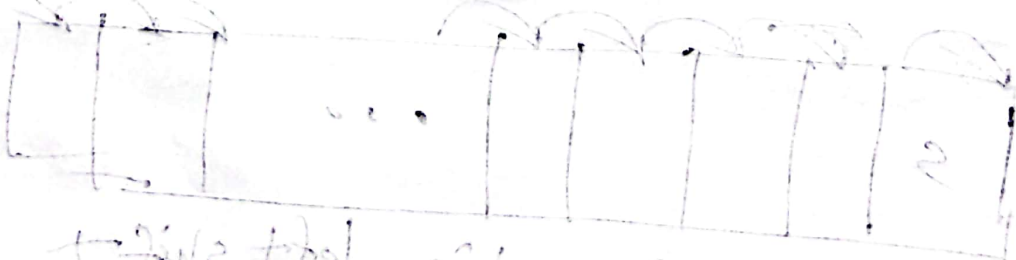
optional

<u>Input</u>	<u>Operation</u>	<u>Result</u>
10100110 x	Logical Right (3 bit)	01010011 00010100
10100110 x	Logical Left(3)	00110000
10100110 x	Arithmetic right(3)	11110100
10100110 x	Arithmetic Left(3)	10110000

8) Multimedia Instruction Set:



MMX



MMX

Chapter - 02

1) Sign Magnitude Representation

The simplest form of representation is the sign-magnitude representation. In this representation, the rightmost bit of the integer is used to indicate the sign of the integer. The remaining bits are used to represent the magnitude of the integer.

$01001000 = +18$
 $10001000 = -18$

There are several advantages and disadvantages of sign-magnitude representation. One advantage is that it is easy to implement. Another advantage is that it allows for easy comparison of numbers. However, it has some disadvantages. For example, it requires a special circuit to add and subtract numbers. Also, it is not as efficient as other representations.

Questions: Give

* Mention data types for multimedia installation of Intel Pentium

2. multimedia product.

* Describe elements of machine instruction

* Illustrate the arithmetic shift operation.

আরও প্রশ্ন

* লক্ষ্য করুন সব Ans. আছে ॥

Chapter - 09

4) Sign Magnitude Representation:

The simplest form of representation that employs a sign bit is the sign magnitude representation. In an n -bit word, the rightmost $n-1$ bits hold the magnitude of the integer.

$$+18 = 00010010$$

$$-18 = 10010010 \text{ (sign magnitude)}$$

There are several drawbacks to sign magnitude representation. One is that addition and subtraction require a consideration of both the signs of the numbers and their relative magnitudes. Another drawback is that there are

two representations of 0.

$$+0_{10} = 00000000$$

$$-0_{10} = 10000000 \text{ (sign m.)}$$

2) Two's Complement Representation:

2's complement representation uses the most significant bit as a sign bit. There is one number of representations of zero. If two numbers with the same sign are added, then overflow occurs if and only if the result has the opposite sign.

3) Addition and Subtraction: (By 2's Complement)

Addition example:

$$\begin{array}{r} 1001 \rightarrow -7 \\ 0101 \rightarrow 5 \\ \hline 1110 \rightarrow -2 \end{array}$$

$$\begin{array}{r} 1100 \rightarrow -4 \\ 0100 \rightarrow 4 \\ \hline 10000 \rightarrow 0 \end{array}$$

Subtract:- To subtract one number from another, take the two's complement of the subtrahend and add it to the minuend.

Ex: $110110 - 10110$
2's complement of 10110 is,

$$\begin{array}{r} 10110 \\ 1 \\ \hline 101001 \end{array}$$

$\therefore$ subtraction =

$$\begin{array}{r} 110110 \\ + 101001 \\ \hline 100000 \end{array}$$

Ans: 100000

Multiplication

Unsigned Integers

$1011 \rightarrow$ Multiplicand
 $\times 1101 \rightarrow$ Multiplier

$$\begin{array}{r} 1011 \\ 0000 \\ 1011 \\ 1011 \\ \hline 1001111 \end{array}$$

Partial Product

1001111 — Product

(সহিত্বন সুলভ নিয়ম)

Algorithm

(i) Pencil and Paper Approach

নিয়ম:

(Multiplier) $\rightarrow M$
 (Multiplicand) $\rightarrow A$

0000 1101 1011

Initialize

(i) এক্ষাণে, $B = 1101$ মানের value 1.
 অর্থাৎ B বর্তমান position 1 হয়, M
 এর সাথে A যোগ করে, A এর
 অঙ্কে রাখতে। $B = 1101$

দ্বিতীয় ধাপ
 এটি বর্তমান
 position. এতে চলবে...

* বিধি মারগ, প্রদত্ত step ৩ ও ৪ চাও।

(ii) C, A, ও এর content গুলোকে right shift করানো। নিচের ২ ঘর right ৫ এর নিচের ক্রমগত ০ বসে যাবে।

(iii) ও এর বর্তমান position ০ হলে, শুধু Right shift করানো।

(iv) M+A করার পর carry bit থাকলে শুধু C এর মান হিসেবে বসে যাবে।

Execution of Example: (Hardware implementation)

C	A	Q	M	Initial Values
0	0000	1101	1011	
0	1011	1101	1011	ADD 1st
0	0101	1110	1011	Shift 1st
0	0010	1111	1011	Shift 2nd
0	1101	1111	1011	ADD 3rd
0	0110	1111	1011	Shift 3rd
1	0001	1111	1011	ADD 4th
0	1000	1111	1011	Shift 4th
0	Now, A = 1000, Q = 1111			
0	Ans: 10001111			

* If the multiplier is negative, straightforward multiplication will not work.

Booth's Algorithm:

A	Q	Q-1	M
0000	0011	0	0111

নিম্নলিখিত:

i) এখানে $Q = 0011$, $Q-1 = 0$, current position Q এবং $Q-1 = 10$

এখানে $Q_i Q_{i-1} = 10$
 $= 00$
 $= 00$

ii) Q_i, Q_{i-1} এর উভয় position Q similar bit হলে, A, Q, $Q-1$ কে Right shift. (Arithmetic Right shift)

iii) Opposite bit হলে
 01 হলে $A = A + M$
 10 হলে $A = A - M$, left shift.

প্রত্যেক বারে $Q-1$ current bit বসে হবে।
 আর Q অবশেষে always 0011.

* (ভাগ বা বিভাগের carry bit রাহে লিখতে হবে)

⑩ Hardware Implementation:

$\begin{array}{c} \overline{2} \\ A \\ 0000 \end{array} \quad \begin{array}{c} \overline{2} \\ Q \\ 0011 \end{array} \quad \begin{array}{c} \overline{1} \\ R-1 \\ 0 \end{array} \quad \begin{array}{c} \overline{2} \\ M \\ 0111 \end{array}$
 Initial Values

$\left\{ \begin{array}{l} 1001 \\ 1100 \\ 1110 \\ 0101 \\ 0010 \end{array} \right\}$
 $\left\{ \begin{array}{l} 0011 \\ 1001 \\ 0100 \\ 0100 \\ 1010 \end{array} \right\}$
 $\left\{ \begin{array}{l} 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{array} \right\}$
 $\left\{ \begin{array}{l} 0111 \\ 0111 \\ 0111 \\ 0111 \\ 0111 \end{array} \right\}$
 Initial Values
 $A \leftarrow A - M$ 1st shift
 $A \leftarrow A + M$ 2nd shift
 $A \leftarrow A + M$ 3rd shift
 $A \leftarrow A + M$ 4th shift
 $\begin{array}{r} 0001 \\ \hline \end{array}$
 $\begin{array}{r} 0101 \\ \hline \end{array}$
 $\begin{array}{r} 00010101 \\ \hline \end{array}$
 Result: 00010101
 $\begin{array}{r} 00010101 \\ \hline \end{array}$
 Result: 00010101

Assignment: Page 302, Booth's algorithm
 এর exampleগুলো রাহে করে
 করা দিতে হবে।
 * আগের example ...

Division

$$\begin{array}{r} 00001101 \rightarrow \text{Quotient} \\ \underline{10010011} \rightarrow \text{Dividend} \end{array}$$

Algorithm

নিয়ম

A
0000

0111

0011

(i)

Normal

Left

Shift করবে প্রথমে

Initial M এর shifted

A

এর sign bit সমান হলে

A - M

অমান না হলে A + M

(ii)

বিভাগ/যোগ

অগুর Result ও পরব

Result এর

sign bit সমান হলে

successful

same না হলে বিভাগ/যোগ

যোগ/বিভাগ Restore করবে।

(iii)

Successful হলে

এর sign bit

0 হলে

$Q_1 = 1$

1 হলে

$Q_1 = 0$

Hardware Implementation: (dividend) (divisor)

A
0000

Q
0111

M = 0011
sign bit

same as
sign bit
0000
1101
0000

1110

1110

0001
1110
0001

1100

1100

0011
0000
0000

1000

1001

0001
1110
0001

0010

0010

Remainder = A = 0001
Quotient = Q (if signs of divisor and dividend same, otherwise 2's of Q)

shift
subtract
restore
1st

shift
subtract
Restore
2nd

shift
subtract
set Q₀ = 1
3rd

shift
subtract
restore
4th

Note

- 6 এর বাইনারি,

১১০

Negative হওয়ার কারণে sign bit 1
স্বাক্ষর করলাম,

1110

2's complement,

0001
1

0010

৬ = 0010

সামনে - 1 দিচ্ছি,

1010

∴ - 6 = 1010

0000

0001

1000

A =

৮ =

2048 — 4(a) → ~~convert~~ convert 8 bit 2
 - ~~convert~~ -

~~2048~~

~~2048~~

* ~~2's complement~~ 2's complement (अस - शिद्धांत)
 बाबा 11 (अस - शिद्धांत)

Algorithm (a) L-8/10.2 (1)

Programmer's manual of 8A bus — 0101 — A
 Programmer's manual of 8A bus — 1100 — A
 Programmer's manual of 8A bus — 1010 — A
 Programmer's manual of 8A bus — 1101 — A
 Programmer's manual of 8A bus — 0110 — A
 Programmer's manual of 8A bus — 1011 — A
 Programmer's manual of 8A bus — 1110 — A
 Programmer's manual of 8A bus — 1111 — A

Chapter-02 and 03

From Mid-term Revise - - -

* আর Computer Evolution এর উপ Assignment
দিখিলো, হাট্ট একবার Revise নিতে হবে।

~~1) The First Generation: Vacuum Tube~~

Questions: (স্বাক্ষরিত)

(1) 2018-1(a)

A hypothetical machine has 6 instructions

A — 1010 — Load AC from memory
0011 — Store AC to memory
0101 — Add to AC from memory
1011 — Sub memory from AC
0110 — Store AC to I/O

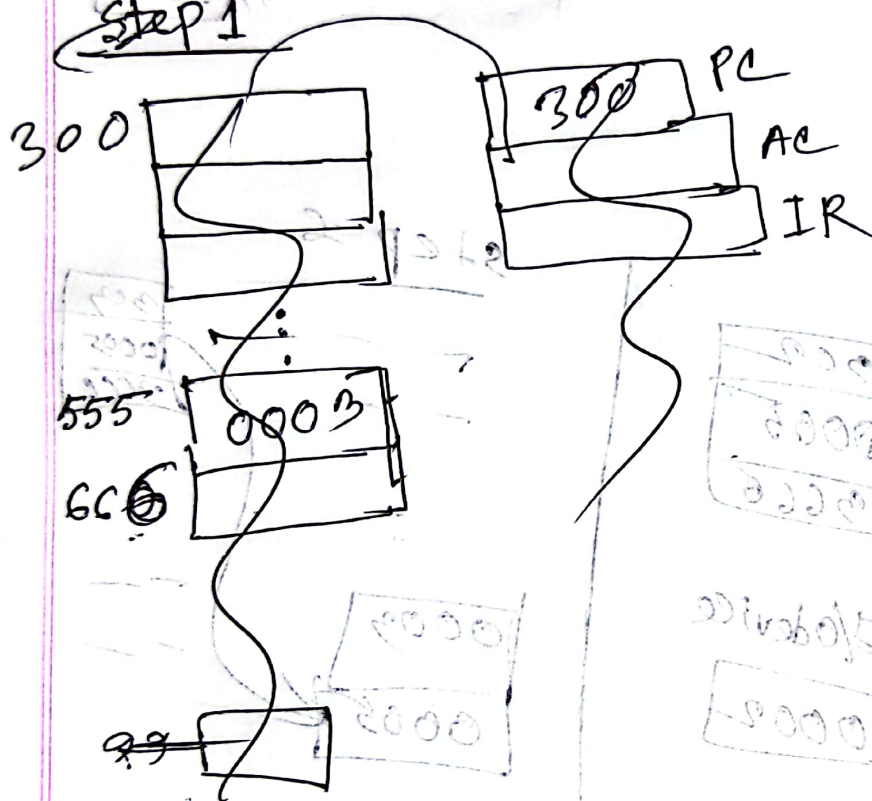
D — 1101 — Load AC from I/O
12 bit identifies I/O device, 4 bit
op code. Show the program execution
using 16 bit instruction
(i) Load AC from device 99
(ii) Add contents of memory
location 555
(iii) Store AC to memory 666.

555 → 3
666 →

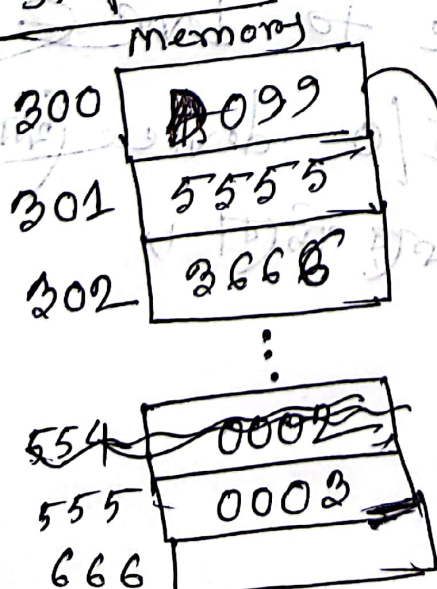
Assume that the next value retrieve from device 99 is 2 and 555 contains a value of 3.

Ans:

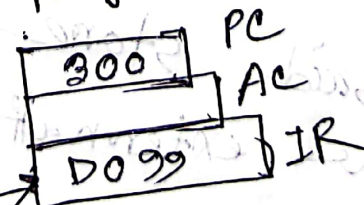
Step 1



Step 1



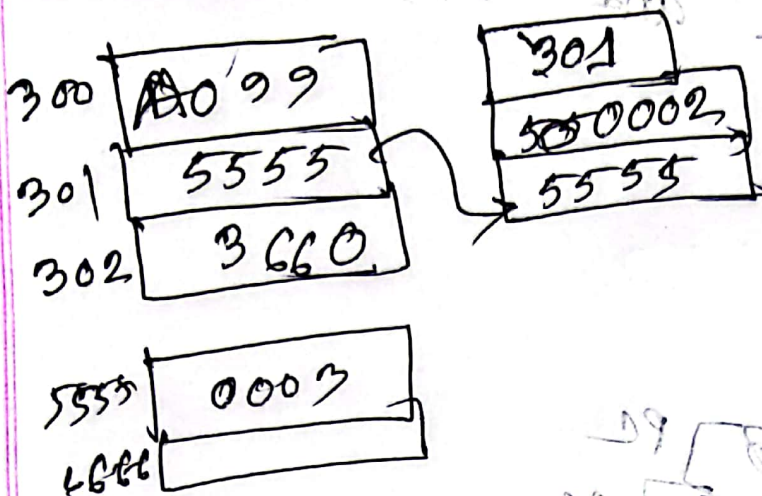
CPU Registers



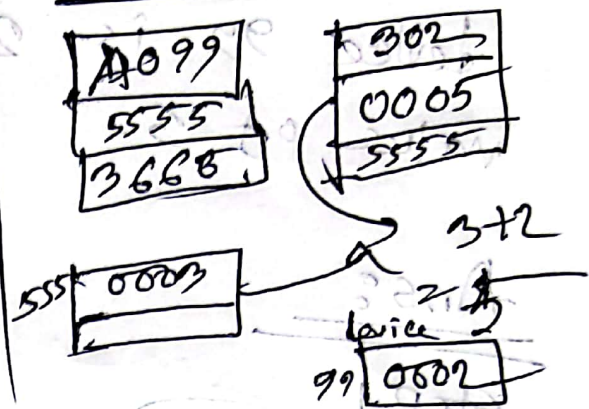
Step 2



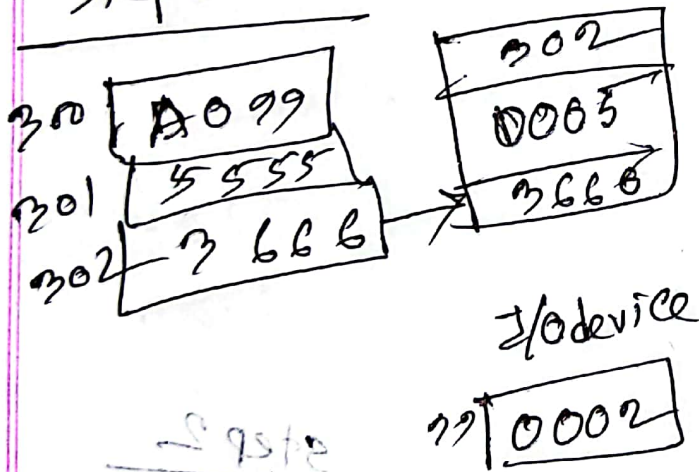
Step 3



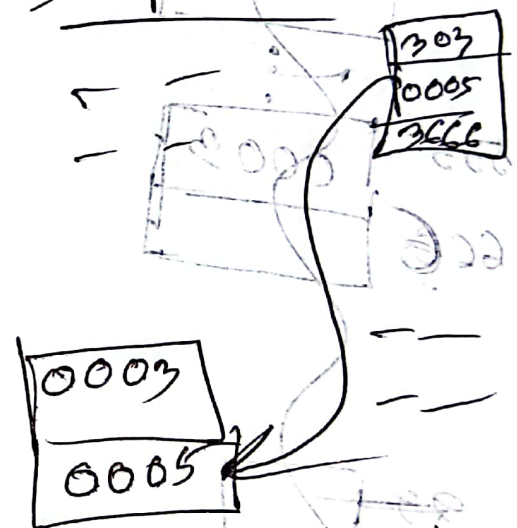
Step 4



Step 5



Step 6



If said store AC to device & the external I/O device (अंतर) ति (store) कहें कि (1)